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Syllabus

Data Visualization Module

Lecture 1: Introduction to the course

Introduction to the Data Visualization module; textbooks; definitions and terminology; visual perception; pre-attentive attributes; Gestalt principles

[slides]

reading material: Chapter 1 (Munzner 2014)

Lecture 2: Nested model

Analysis framework: nested model; data abstraction (what); common types of data; task abstraction (why)

[slides]

reading material: Chapter 2,3,4 (Munzner 2014)

resources: Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 2]; [video Chapter 3]; [video Chapter 4]

Lecture 3: Visual encoding

Visual encoding; marks and channels; color in visualization; color palette; color deficiency; color spaces

[slides]

reading material: Chapter 5: Marks and Channels, Chapter 10: Map Color and Other Channels (Munzner 2014)

resources: VizPalette **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 5-a]; [video Chapter 5-b] [video Chapter 10-a]; [video Chapter 10-b]; [video Chapter 10-c]

Lecture 4: Common charts

Visualize tabular data; common visual idioms and charts; scatterplot; (stacked) bar chart; streamgraph; dot/line chart; Gantt chart; slopegraph; heatmap; radial bar chart; star plot; radar plot; pie chart; coxcomb chart; parallel coordinates; dual-axis charts; Visual vocabulary

[slides]

reading material: Chapter 7: Arrange Tables (Munzner 2014)

resources: Visual Vocabulary **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 7-a]; [video Chapter 7-b]

Lecture 5: Static plotting in Python

Static plotting in Python; basic plotting in matplotlib; style and personalize plots; towards more advanced plotting in matplotlib; introduction to seaborn; rules of thumb in data visualization

[slides-a] [slides-b]

tutorials: - 01-matplotlib/01-figures_subplots.ipynb — figures and subplots - 01-matplotlib/02-plotting.ipynb — basic plotting - 01-matplotlib/03-customizing_appearance.ipynb — style and customisation - 01-matplotlib/04-visual-encoding.ipynb — visual encoding: marks, channels, and color - 01-matplotlib/05-advanced_subplotting.ipynb — advanced subplotting - 01-matplotlib/06-charts-matplotlib.ipynb — common charts (scatter, bar, line, histogram) with Gapminder

resources:

- matplotlib: Examples | Tutorial | User guide
- seaborn: Gallery | Tutorial | API

Lecture 6: Geographical plotting in Python

The importance of spatial thinking; exploring spatial phenomena visually; how to create informative thematic cartography; main thematic cartography types and rules of thumbs; pitfalls of using spatial data in computational disciplines

[slides]

Lecture 7: Visualize static networks in Python

Visualize static networks using NetworkX and Matplotlib

Lecture 8: Interactive Views

How to handle complexity; manipulate: change, select, navigate; facet: juxtapose, partition, superimpose; reduce: filter, aggregate, embed

[slides]

tutorials: - 02-interactive/01-plotly-express.ipynb — interactive charts with Plotly Express (same 4 charts + animated bubble chart) - 02-interactive/02-altair.ipynb — Grammar of Graphics with Altair (linked selections, layering)

resources: Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 11-12a] [video Chapter 11-12b] [video Chapter 13] [video Chapter 14]

Lecture 9: Introduction to D3.js

Why D3, why now: when declarative grammars run out; the web platform in 10 minutes (HTML, CSS, JavaScript, the DOM, SVG); SVG attributes as visual channels; D3 in three ideas (selections, scales, joins); the margin convention; building the Gapminder scatterplot end-to-end; Rosetta stone — the same chart in matplotlib, Altair, and D3

[slides]

tutorials: - 09-d3js-intro/09a-building-svg-with-d3.html — HTML scaffolding, `select/append/attr/style`, SVG element catalog, coordinate system, attribute playground - 09-d3js-intro/09b-data-driven-svg.html — the data join (`selectAll · data · join`); explicit enter / update / exit decomposition - 09-d3js-intro/09c-scales-and-axes.html — scales (linear / log / band toggle), margin convention, axes; same skeleton on Gapminder - 09-d3js-intro/09d-loading-data.html — `d3.csv` accessor; helpers (`extent`, `group`, `rollup`); formatters (`d3.format`, `d3.timeFormat`) - 09-d3js-intro/09e-first-scatterplot.html — capstone: full Gapminder scatterplot end-to-end

reading material: - Murray, *Interactive Data Visualization for the Web* (2nd ed.), chapters 1–6 — free online - Bostock, “Thinking with Joins”

resources: d3js.org | Observable D3 collection

Legacy slide decks `dv09-D3-course-lesson-1.pdf` and `dv09-D3-course-lesson-2.pdf` (third-party material) are kept under `slides/` as supplementary reference.

Lecture 10: D3 Shapes, Layouts, and Interaction

SVG path mini-language; D3 path generators (`d3.line`, `d3.area`, curve types); bar charts and `scaleBand`; stacked bars with `d3.stack`; the General Update Pattern and `selection.join`; key functions vs index joins; custom join with three callbacks; transitions (duration, delay, easing, chained, staggered); event listeners and the event-state-render loop; interactive Gapminder capstone

[slides]

tutorials: - `10-d3js-shapes-layouts/10a-svg-paths-and-line.html` —

SVG path mini-language; `d3.line`, `d3.area`; curves - `10-d3js-shapes-layouts/10b-bar-charts-and-stack.`

— bar charts; `scaleBand`; `d3.stack` layout - `10-d3js-shapes-layouts/10c-update-pattern.html`

— General Update Pattern; `selection.join`; key functions - `10-d3js-shapes-layouts/10d-transitions.htm`

— transitions, easing, chaining, staggered delays - `10-d3js-shapes-layouts/10e-interactions-capstone.htm`

— interactive Gapminder scatterplot (hover, click, year slider)

Legacy slide decks `dv10-D3-course-lesson-3.pdf` and `dv10-D3-course-lesson-4.pdf` (third-party material) are kept under `slides/` as supplementary reference.

Lecture 11: Interactions and Event Listeners

Data structures and visualization structures; interaction and event listeners; thinking in D3; how to keep learning

[slides]

Lecture 12: Graphs Visualization

Interactive graph visualization; advanced examples

[tutorial]

References

Munzner, T. 2014. *Visualization Analysis and Design*. AK Peters Visualization Series. CRC Press. <https://books.google.it/books?id=dznSBQAAQBAJ>.